22AM034-BUSINESS INTELLIGENCE

UNIT 1 & LP1 - EFFECTIVE AND TIMELY DECISIONS

1. BUSINESS INTELLIGENCE DEFINITION

The main purpose of business intelligence systems is to provide knowledge workers with tools and methodologies that allow them to make decisions at the right time.

1.1 What is Business Intelligence ?

Business Intelligence (BI) is about getting the right information, to the right decision makers, at the right time.BI is an enterprise-wide platform that supports reporting, analysis and decision making.BI leads to fact-based decision making.

1.2 Definition :

Business intelligence may be defined as a set of mathematical models and analysis methodologies that exploit the available data to generate information and knowledge useful for complex decision-making processes. Loads of heterogeneous data available everywhere. It is possible to convert such data into information and knowledge that can then be used by decision makers. Business intelligence may be defined as a set of mathematical models and analysis methodologies that exploit the available data to generate information and knowledge useful for complex decision-making processes. The main purpose of business intelligence systems is to provide knowledge workers with tools and methodologies that allow them to make effective and timely decisions.

1.3 BUSINESS INTELLIGENCE PROCESS

1.3.1 The BI Process

The business intelligence is utilized in different ways and for different purposes by individual companies, the process is uniform throughout all industries and typically unfolds as follows:

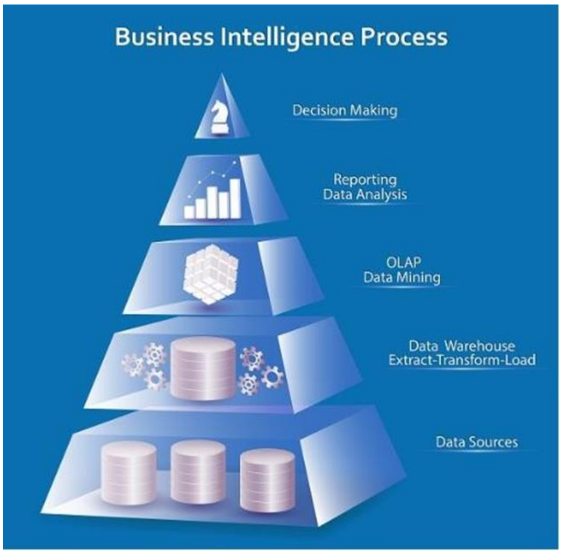
- Data from various sources -including internal company data and external market data - is collected, integrated, and then stored; because “big data” is commonly used, data is commonly stored in what is called a data warehouse, created by a data engineer

- Data sets are created and prepared for data analysis, often by creating data analysis models

- Data analysts run queries against the data sets or models

- the results of queries are used to produce visualizations in the form of charts, graphs, histograms, or other visual representations, along with BI dashboards and reports

- Decision-makers utilize the data visualizations and reports to help them in making decisions; they may also use their BI dashboard to probe further into the data for more information.



Business Intelligence Process

1.3.2 How does the business intelligence process work?

A business intelligence architecture includes more than just BI software. Business intelligence data is typically stored in a data warehouse built for an entire organization or in smaller data marts that hold subsets of business information for individual departments and business units, often with ties to an enterprise data warehouse. In addition, data lakes based on Hadoop clusters or other big data systems are increasingly used as repositories or landing pads for BI and analytics data, especially for log files, sensor data, text and other types of unstructured or semi structured data.

BI data can include historical information and real-time data gathered from source systems as it's generated, enabling BI tools to support both strategic and tactical decision-making processes. Before it's used in BI applications, raw data from different source systems generally must be integrated, consolidated and cleansed using data integration and data quality management tools to ensure that BI teams and business users are analyzing accurate and consistent information.

From there, the steps in the BI process include the following:

- Data preparation, in which data sets are organized and modeled for analysis;
- Analytical querying of the prepared data;
- Distribution of key performance indicators (KPIs) and other findings to business users; and
- Use of the information to help influence and drive business decisions.

Initially, BI tools were primarily used by BI and IT professionals who ran queries and produced dashboards and reports for business users. Increasingly, however, business analysts, executives and workers are using business intelligence platforms themselves, thanks to the development of self-service BI and data discovery tools. Self-service business intelligence environments enable business users to query BI data, create data visualizations and design dashboards on their own.

BI programs often incorporate forms of advanced analytics, such as data mining, predictive analytics, text mining, statistical analysis, and big data analytics. A common example is predictive modeling that enables what-if analysis of different business scenarios. In most cases, though, advanced analytics projects are conducted by separate teams of data scientists, statisticians, predictive modelers, and other skilled analytics professionals, while BI teams oversee more straightforward querying and analysis of business data.

1.4 Why is business intelligence important?

The role of business intelligence is to improve an organization's business operations using relevant data. Companies that effectively employ BI tools and techniques can translate their collected data into valuable insights about their business processes and strategies. Such insights can then be used to make better business decisions that increase productivity and revenue, leading to accelerated business growth and higher profits.

Without BI, organizations cannot readily take advantage of data-driven decision-making. Instead, executives and workers are primarily left to base important business decisions on other factors, such as accumulated knowledge, previous experiences, intuition, and gut feelings. While those methods can result in good decisions, they are also fraught with the potential for errors and missteps because of the lack of data underpinning them.

1.5 Benefits of business intelligence

A successful BI program produces a variety of business benefits in an organization. For example, BI enables C-suite executives and department managers to monitor business performance on an ongoing basis so they can act quickly when issues or opportunities arise. Analysing customer data helps make marketing, sales and customer service efforts more effective. Supply chain, manufacturing and distribution bottlenecks can be detected before they cause financial harm. HR managers are better able to monitor employee productivity, labour costs and other workforce data.

Overall, the key benefits that businesses can get from BI applications include the ability to:

- speed up and improve decision-making;
- optimizes internal business processes;

- increases operational efficiency and productivity;
- spot business problems that need to be addressed;
- identifies emerging business and market trends;
- develops stronger business strategies;
- drive higher sales and new revenues; and
- gains a competitive edge over rival companies.

BI initiatives also provide narrower business benefits among them, making it easier for project managers to track the status of business projects and for organizations to gather competitive intelligence on their rivals. In addition, BI, data management and IT teams themselves benefit from business intelligence, using it to analyze various aspects of technology and analytics operations.

1.5 Types of business intelligence tools and applications

Business intelligence combines a broad set of data analysis applications designed to meet different information needs. Most are supported by both self-service BI software and traditional BI platforms. The list of BI technologies that are available to organizations includes the following:

- Ad hoc analysis. Also known as ad hoc querying, this is one of the foundational elements of modern BI applications and a key feature of self-service BI tools. It's the process of writing and running queries to analyze specific business issues. While ad hoc queries are typically created on the fly, they often end up being run regularly, with the analytics results incorporated into dashboards and reports.

- Online analytical processing (OLAP). One of the early BI technologies, OLAP tools enable users to analyze data along multiple dimensions, which is particularly suited to complex queries and calculations. In the past, the data had to be extracted from a data warehouse and stored in multidimensional OLAP cubes, but it's increasingly possible to run OLAP analysis directly against columnar databases.

- Mobile BI. Mobile business intelligence makes BI applications and dashboards available on smartphones and tablets. Often used more to view data than to analyze it, mobile BI tools typically are designed with an emphasis on ease of use. For example, mobile dashboards may only display two or three data visualizations and KPIs so they can easily be viewed on a device's screen.

- Real-time BI. In real-time BI applications, data is analyzed as it's created, collected and processed to give users an up-to-date view of business operations, customer behavior, financial markets and other areas of interest. The real-time analytics process often involves streaming data and supports decision analytics uses, such as credit scoring, stock trading and targeted promotional offers.

- Operational intelligence (OI). Also called operational BI, this is a form of real-time analytics that delivers information to managers and frontline workers in business operations. OI applications are designed to aid in operational decision-making and enable faster action on issues -- for example, helping call center agents to resolve problems for customers and logistics managers to ease distribution bottlenecks.

- Software-as-a-service BI. SaaS BI tools use cloud computing systems hosted by vendors to deliver data analysis capabilities to users in the form of a service that's typically priced on a subscription basis. Also known as cloud BI, the SaaS option increasingly offers multi-cloud support, which enables organizations to deploy BI applications on different cloud platforms to meet user needs and avoid vendor lock-in.

- Open-source BI (OSBI). Business intelligence software that is open source typically includes two versions: a community edition that can be used free of charge and a subscription-based commercial release with technical support by the vendor. BI teams can also access the source code for development uses. In addition, some vendors of proprietary BI tools offer free editions, primarily for individual users.

- Embedded BI. Embedded business intelligence tools put BI and data visualization functionality directly into business applications. That enables business users to analyze data within the applications they use to do their job. Embedded analytics features are most incorporated by application software vendors, but corporate software developers can also include them in homegrown applications.

- Collaborative BI. This is more of a process than a specific technology. It involves the combination of BI applications and collaboration tools to enable different users to work together on data analysis and share information with one another. For example, users can annotate BI data and analytics results with comments, questions and highlighting via the use of online chat and discussion tools.

- Location intelligence (LI). This is a specialized form of BI that enables users to analyze location and geospatial data, with map-based data visualization functionality incorporated. Location intelligence offers insights on geographic elements in business data and operations. Potential uses include site selection for retail stores and corporate facilities, location-based marketing, and logistics management.

2. EFFECTIVE AND TIMELY DECISIONS

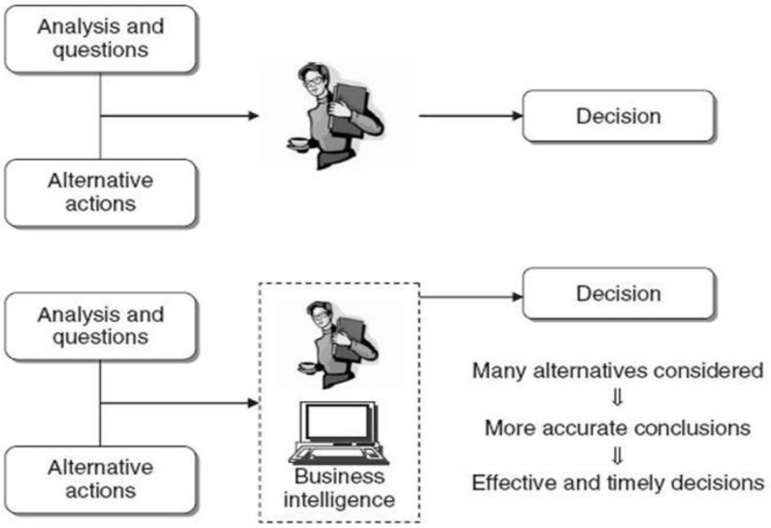
- The main purpose of business intelligence systems is to provide knowledge workers with tools and methodologies that allow them to make effective and timely decisions.
- In complex organizations, public or private, decisions are made on a continual basis. Such decisions may be critical, have long- or short- term effects and involve people and roles at various hierarchical levels.
- The ability of these knowledge workers to make decisions, both as individuals and as a community, is one of the primary factors that influence the performance and competitive strength of a given organization

2.1 Effective Decisions

- The application of analytical methods allows decision makers to rely on information and knowledge which are more dependable.
- As a result, they can make better decisions and devise action plans that allow their objectives to be reached in a more effective way.

2.2 Timely Decisions

- Enterprises operate in economic environments characterized by growing levels of competition and high dynamism.
- Therefore, the ability to rapidly react to the actions of competitors and to new market conditions is a critical factor in the success or even the survival of a company.



Benefits of a Business Intelligence System

DISCUSSION QUESTIONS:

1. How does data quality impact the effectiveness of decision-making in business intelligence?
2. What role does real-time data play in enhancing timely decision-making processes?
3. In what ways can organizations balance the need for thorough analysis with the urgency of timely decisions?
4. How can technology, such as AI and machine learning, improve decision-making speed and accuracy in business intelligence?
5. What are some common pitfalls organizations face when making decisions based on business intelligence, and how can they be mitigated?